



Abhijeet Vaibhav
Electrical Engineering
Indian Institute of Technology Bombay
Specialization: Control and Computing

21D070004
Dual Degree (B.Tech. + M.Tech.)
Gender: Male
DOB: 09/01/2003

Examination	University	Institute	Year
Graduation	IIT Bombay	IIT Bombay	2026

Projects Undertaken

Legal Section Prediction AI Model from Case Descriptions

(September 2025- October 2025)

Self

- Created and preprocessed a custom legal sections dataset from JSON to enable semantic search. Implemented a semantic search system using SentenceTransformer (all-MiniLM-L6-v2) for embedding legal text sections.
- Developed a Streamlit-based UI allowing users to query case descriptions or section numbers and retrieve relevant legal sections.
- Applied sentence tokenization and dynamic thresholding on cosine similarity scores for precise multi-sentence semantic matching.

Developed NER model using CoNLL dataset for entity extraction

(October 2025)

Course Project: EE782 – Advanced ML

- Developed a Named Entity Recognition (NER) system using the CoNLL dataset to accurately identify and classify entities in text.
- Implemented sequence labelling techniques to classify entities in diverse text corpora efficiently.

Hostel Room Security Using AI Face Recognition and Voice Alerts

(September 2025)

Course Project: EE782 – Advanced ML

- Developed an AI-based hostel room security system using face recognition and speech transcription to detect and deter unknown visitors.
- Integrated DeepFace Facenet and OpenAI Whisper models within cloud constraints for reliable identification and transcription.
- Ensured effective monitoring using video/audio processing despite hardware limitations, achieving high accuracy in detection and escalation.
- Implemented conditional voice alerts via Google Text-to-Speech triggered only when unauthorized individuals are present without the owner.

BANK OF BARODA — ATM FAILURE PREDICTION

(July 2025- June 2026)

DDP Guide: Prof. Debraj Chakraborty

- Will be developing a data integration pipeline to combine and preprocess ATM fault ticket data using Python (pandas, NumPy, datetime).
- Plan to engineer time-based features for fault pattern analysis and predictive modeling using XGBoost.
- Intend to apply Random Under-Sampling to handle class imbalance and improve model robustness.
- Will evaluate model performance through classification reports, ROC-AUC scores, and confusion matrices.

HEART FAILURE DETECTION

(June-July 2025)

Self

- Built a machine learning model to predict heart failure risk from patient clinical data.
- Performed EDA, feature engineering, and hyperparameter tuning (GridSearchCV) to optimize model performance.
- Achieved approx. 87% accuracy and high ROC-AUC with XGBoost; used SHAP for model interpretability and deployment-ready model saving.

SNAKE ROBOT

(August - November 2024)

SRE Guide: Prof. Debraj Chakraborty

- Developed an automated Python pipeline using Sobel and Laplacian variance methods to extract sharp frames from robot-mounted cameras navigating confined spaces.
- Implemented threshold-based filtering for high-quality frame selection, enabling precise spatial measurements and efficient processing of large video datasets.
- Developed a computer vision pipeline using OpenCV to detect and extract cave walls contours from snake robot camera frames through morphological processing and thresholding techniques.
- Implemented bounding box analysis and visual annotations to measure and display 2D spatial dimensions, enabling precise size estimation in inaccessible environments.

BIOMETRICS

(August - Novemeber 2024)

Course Project: EE 678 - Wavelets

- Built RTPS deformation-correction and Sequential CNN pipelines to match contactless and contact-based fingerprints.
- Applied DWT and Adaptive Histogram Equalization for ridge enhancement and robust minutiae extraction.
- Achieved 83% accuracy with CNN and 74% with RTPS on approx.3,120 fingerprint images.

High Sensitivity Optical Detection System for Spectrometry (January - March 2024) Course Project: EDL

- Designed and built a portable high-sensitivity spectrometer for accurate wavelength-intensity measurement.
- Developed custom PCB and lock-in amplifier for low-noise photodiode signal acquisition.
- Integrated optical, electronic, and mechanical components using 3D printing and laser cutting
- Validated system performance through absorption and transmission spectrum measurements.

Technical Skills

Programming C++, VHDL, Python, Julia

Tools and libraries L^AT_EX, MATLAB

PoR & Extracurriculars

Teaching Assistant— EE 204: Analog Circuits

(2025- Autumn)

- Taken attendance and maintained records.
- Assisted in conducting and demonstrating in-class analog circuit experiments.
- Conducted tutorials and clarified student doubts.
- Prepared, administered, and graded exams.

Mess Secretary— Hostel 3

(2022-23)

- Coordinated daily mess operations and meal services.
- Communicated between students, staff, and administration.
- Managed menu planning and upheld food quality and hygiene.
- Organized special mess events.